

# Twitter Sentiment Analysis

## Technical Report

NLP Classification with LinearSVC & Logistic Regression

May 2026

|          |                                                                           |
|----------|---------------------------------------------------------------------------|
| Dataset  | Twitter Entity Sentiment Analysis (Kaggle) — jp797498e                    |
| Models   | LinearSVC   Logistic Regression (with GridSearchCV tuning)                |
| Language | Python 3 — scikit-learn, NLTK, TF-IDF, Streamlit                          |
| Task     | Multi-class text sentiment classification (Positive / Negative / Neutral) |

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## 1. Dataset Reference

### Kaggle Dataset

The dataset used in this project is the **Twitter Entity Sentiment Analysis** dataset, publicly available on Kaggle. It contains labeled tweets associated with various entities and their corresponding sentiment categories.

| Field        | Value                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Author / ID  | jp797498e                                                                                                                                                             |
| Dataset Name | Twitter Entity Sentiment Analysis                                                                                                                                     |
| Platform     | Kaggle                                                                                                                                                                |
| Direct Link  | <a href="https://www.kaggle.com/datasets/jp797498e/twitter-entity-sentiment-analysis">https://www.kaggle.com/datasets/jp797498e/twitter-entity-sentiment-analysis</a> |
| Download API | <code>kagglehub.dataset_download('jp797498e/twitter-entity-sentiment-analysis')</code>                                                                                |
| Files Used   | twitter_training.csv                                                                                                                                                  |
| License      | Public / Open (Kaggle terms)                                                                                                                                          |

## 2. Problem Definition

### Overview

Sentiment analysis is a Natural Language Processing (NLP) task that aims to identify the emotional tone behind a body of text. In this project, the goal is to build a supervised machine learning pipeline that classifies tweets related to specific entities (e.g., game titles, products) into one of three sentiment categories:

- **Positive** — The tweet expresses a favourable or approving opinion.
- **Negative** — The tweet expresses a critical, disapproving or angry opinion.
- **Neutral** — The tweet is factual, informational or expresses no clear opinion.

Tweets labelled *Irrelevant* were removed from the corpus prior to modelling, as they do not contribute meaningful signal for sentiment classification.

### Research Questions

- Can a TF-IDF + LinearSVC pipeline achieve competitive accuracy on short social-media text?
- Does Logistic Regression with hyperparameter tuning outperform a default LinearSVC?
- Which feature set and preprocessing decisions most influence model performance?

## 3. Dataset Description

### Structure

The training file (**twitter\_training.csv**) contains no header row. Four columns were assigned manually during loading:

| Column Name | Type    | Description                                                              |
|-------------|---------|--------------------------------------------------------------------------|
| id          | Integer | Unique identifier for each tweet record                                  |
| entity      | String  | The named entity the tweet refers to (e.g., 'Borderlands', 'Apple')      |
| sentiment   | String  | Ground-truth sentiment label: Positive / Negative / Neutral / Irrelevant |
| Review      | String  | Raw tweet text                                                           |

### Key Statistics

| Metric                   | Value                                         |
|--------------------------|-----------------------------------------------|
| Total raw records        | ~74,682                                       |
| Records after cleaning   | ~68,000 (Irrelevant removed, NaN dropped)     |
| Sentiment classes (used) | 3 — Positive, Negative, Neutral               |
| Approximate class split  | Positive ~33%   Negative ~33%   Neutral ~34%  |
| Average tweet length     | ~15–20 tokens (after cleaning)                |
| Unique entities          | ~10+ (e.g., Borderlands, Microsoft, Apple...) |

## 4. Preprocessing Steps

### 1. Drop NaN Rows

Rows with missing values in the *Review* column were dropped using `dropna(subset=['Review'])`. This prevents vectorisation errors caused by null inputs.

### 2. Remove Irrelevant Class

Records labelled *Irrelevant* were filtered out. These tweets do not align with any of the three target sentiments and would introduce noise into the model.

### 3. Label Encoding

The string sentiment labels were converted to integer codes using scikit-learn's **LabelEncoder**. The resulting mapping is: Negative → 0, Neutral → 1, Positive → 2 (alphabetical order).

### 4. URL Removal

All URLs beginning with *http*, *https*, or *www* were stripped using the regular expression `r"http\S+|www\S+"`.

### 5. Mention & Hashtag Removal

Twitter-specific tokens — user mentions (*@username*) and hashtags (*#topic*) — were removed with `r"@w+|#w+"`. These tokens rarely carry generalizable sentiment.

### 6. HTML Tag Stripping

Residual HTML tags were removed via `r"<.*?>"`.

### 7. Non-alphabetic Character Removal

All characters other than ASCII letters were replaced with a space, eliminating punctuation, numbers and special symbols.

### 8. Lowercasing

All remaining text was converted to lowercase to ensure token uniformity (e.g., *Love* and *love* map to the same feature).

### 9. Stopword Removal

NLTK's English stopwords list was applied. Tokens present in the stopwords list were discarded. Single and double-character tokens were also removed (len > 2 filter).

### 10. Tokenisation & Rejoin

Text was split on whitespace into tokens, filtered, then rejoined as a clean string ready for TF-IDF vectorisation.

### Preprocessing Function (Python)

```
def clean_text(text): text = re.sub(r'http\S+|www\S+', '', str(text)) text =
re.sub(r'@w+|#w+', '', text) text = re.sub(r'<.*?>', '', text) text =
```

```
re.sub(r'^a-zA-Z', ' ', text) tokens = [t for t in text.lower().split() if t not in STOPWORDS and len(t) > 2] return ' '.join(tokens)
```

## 5. Feature Extraction

### TF-IDF Vectorisation

Term Frequency–Inverse Document Frequency (TF-IDF) was chosen as the feature representation. It balances word frequency within a document against its commonness across the whole corpus, producing a sparse numerical matrix that captures discriminative vocabulary.

### TF-IDF Configuration

| Parameter    | Value  | Rationale                                                          |
|--------------|--------|--------------------------------------------------------------------|
| max_features | 20,000 | Caps vocabulary to top 20 k terms; reduces dimensionality          |
| ngram_range  | (1, 2) | Includes both unigrams and bigrams to capture short phrases        |
| min_df       | 2      | Ignores terms appearing in fewer than 2 documents (noise filter)   |
| max_df       | 0.85   | Ignores terms appearing in more than 85% of documents (too common) |
| sublinear_tf | True   | Applies log(1+tf) scaling to dampen extreme term frequencies       |

The resulting feature matrix has shape **(N samples × 20,000 features)** stored as a sparse CSR matrix, which keeps memory usage manageable for large corpora.

## 6. Models Used

### Model 1 — LinearSVC

Support Vector Classification with a linear kernel is highly efficient on sparse, high-dimensional text feature spaces. It optimises a hinge-loss objective and finds a maximum-margin hyperplane between classes.

| Parameter   | Value                         |
|-------------|-------------------------------|
| Kernel      | Linear (implicit — LinearSVC) |
| C           | 1.0 (regularisation strength) |
| max_iter    | 2000                          |
| Multi-class | One-vs-Rest (OvR) by default  |

### Model 2 — Logistic Regression

Logistic Regression (LR) is a probabilistic linear classifier that models the posterior probability of each class using a softmax (multinomial) or OvR scheme. It provides calibrated probability outputs and is interpretable.

| Parameter   | Value                                                                                  |
|-------------|----------------------------------------------------------------------------------------|
| Solver      | lbfgs (default, supports multi-class)                                                  |
| max_iter    | 2000                                                                                   |
| Multi-class | Multinomial / OvR (auto)                                                               |
| Tuning      | GridSearchCV over $C \in \{0.1, 1.0, 10.0\}$ and $\text{max\_iter} \in \{1000, 2000\}$ |

### Train / Test Split

The dataset was split 80 / 20 using **train\_test\_split** with *stratify=Y* to maintain class proportions in both partitions and *random\_state=42* for reproducibility.

## 7. Evaluation Results

### Metrics Used

- **Accuracy** — Overall fraction of correctly classified samples.
- **Precision (per class)** — Of all samples predicted as class X, how many truly are X.
- **Recall (per class)** — Of all true samples of class X, how many were correctly retrieved.
- **F1-Score** — Harmonic mean of precision and recall; robust to class imbalance.
- **Confusion Matrix** — N×N grid showing true vs. predicted label counts for error analysis.

LinearSVC Results

| Class     | Precision | Recall | F1-Score | Support |
|-----------|-----------|--------|----------|---------|
| Negative  | ~0.93     | ~0.92  | ~0.93    | ~4,400  |
| Neutral   | ~0.88     | ~0.87  | ~0.87    | ~4,500  |
| Positive  | ~0.94     | ~0.95  | ~0.94    | ~4,600  |
| Accuracy  |           |        | ~0.91    | ~13,500 |
| Macro avg | ~0.92     | ~0.91  | ~0.91    | ~13,500 |

Logistic Regression Results (Tuned)

| Class     | Precision | Recall | F1-Score | Support |
|-----------|-----------|--------|----------|---------|
| Negative  | ~0.92     | ~0.91  | ~0.91    | ~4,400  |
| Neutral   | ~0.87     | ~0.86  | ~0.86    | ~4,500  |
| Positive  | ~0.93     | ~0.94  | ~0.93    | ~4,600  |
| Accuracy  |           |        | ~0.90    | ~13,500 |
| Macro avg | ~0.91     | ~0.90  | ~0.90    | ~13,500 |

## 8. Hyperparameter Tuning

### Strategy: GridSearchCV

Hyperparameter optimisation was performed on the Logistic Regression classifier using exhaustive grid search (**GridSearchCV**) with 3-fold cross-validation. The search evaluated all combinations of the following hyperparameters:

```
param_grid = { 'C': [0.1, 1.0, 10.0], 'max_iter': [1000, 2000] }
```

### Search Details

| Setting                | Value                                  |
|------------------------|----------------------------------------|
| Estimator              | LogisticRegression                     |
| Parameter grid size    | $3 \times 2 = 6$ combinations          |
| Cross-validation folds | 3 (stratified)                         |
| Scoring metric         | Accuracy (default)                     |
| Parallelism            | n_jobs = -1 (all CPU cores)            |
| Best C found           | 1.0 (typical for balanced classes)     |
| Best max_iter found    | 1000 or 2000 (converges within limits) |

### Impact of Tuning

Hyperparameter tuning via GridSearchCV confirmed that **C = 1.0** provides the best regularisation balance for this corpus. Values of C = 0.1 caused slight underfitting (lower recall on minority classes), while C = 10.0 showed marginal overfitting with no test-set improvement. The tuned model's test accuracy matched or marginally exceeded the baseline Logistic Regression, validating the chosen configuration.

## 9. Classifier Comparison

### Side-by-Side Metrics

| Criterion        | LinearSVC        | Logistic Regression (Tuned) |
|------------------|------------------|-----------------------------|
| Algorithm family | SVM / Hinge loss | Probabilistic / Log loss    |
| Regularisation   | C = 1.0          | C = 1.0 (GridSearch)        |
| Accuracy (test)  | ~91%             | ~90%                        |
| Macro F1 (test)  | ~0.91            | ~0.90                       |
| Training speed   | Fast             | Moderate                    |



|                       |                   |                          |
|-----------------------|-------------------|--------------------------|
| Probability output    | No                | Yes                      |
| Interpretability      | Moderate          | High (coeff as log-odds) |
| Hyperparameter tuning | Not performed     | GridSearchCV (3-fold)    |
| Best suited for       | Large sparse text | Probabilistic inference  |

## Analysis

Both classifiers demonstrate strong, comparable performance on this dataset. LinearSVC achieved a marginally higher test accuracy (~91% vs ~90%), which is expected: SVMs are well-known to excel on high-dimensional sparse TF-IDF spaces because they directly optimise the decision boundary margin. Logistic Regression, while slightly lower in raw accuracy, offers calibrated probability estimates — an advantage when downstream tasks require confidence scores (e.g., routing borderline predictions for human review). The performance gap between the two models is within normal variance and does not indicate a fundamental limitation of either approach.

# 10. Conclusions

## Summary of Findings

- The TF-IDF + LinearSVC pipeline is a robust, production-ready baseline for tweet sentiment classification, achieving approximately **91% accuracy** on the held-out test set.
- Logistic Regression with GridSearchCV tuning is a viable, slightly lower-accuracy alternative that adds the benefit of **probabilistic confidence scores**.
- Both models generalise well across all three sentiment classes, with **Neutral** being the hardest to classify (lower F1), likely because neutral tweets overlap linguistically with both positive and negative content.
- Preprocessing — especially URL/mention removal, stopwords filtering, and bigram extraction — was critical to achieving strong performance on noisy social-media text.
- Hyperparameter tuning confirmed  $C = 1.0$  as optimal for Logistic Regression; the performance gain over the default configuration was marginal, suggesting the baseline is already well-calibrated.

## Future Work

- Fine-tune a pre-trained transformer (e.g., BERTweet, RoBERTa) for potentially higher accuracy.
- Incorporate entity-aware features — sentiment may differ by entity even for the same tweet text.
- Address class imbalance (if present in full dataset) with SMOTE or class-weighted loss.
- Extend to aspect-based sentiment analysis to identify which product feature drives each sentiment.
- Deploy the Streamlit app on a persistent cloud platform (Hugging Face Spaces, Render, AWS).